PRACTICAL: 11

Student Management CRUD with Search and Filter

Objective:

To create a student management module in PHP and MySQL that can add, view, edit, delete, search and filter student records. All database operations use MySQLi prepared statements, and every operation shows a clear success or failure message.

Theory:

1. **CRUD.** CRUD stands for Create, Read, Update and Delete. These are the four basic operations on stored data. In this practical they are done with the SQL commands INSERT, SELECT, UPDATE and DELETE on the students table of the collage1 database.
2. **MySQLi prepared statements.** MySQLi is the improved MySQL extension of PHP. A prepared statement is written with ? placeholders, using prepare(). The values are attached with bind_param() and the query runs with execute(). Data is sent separately from the SQL text, so SQL injection is prevented.
3. **Validation and safe output.** The form values are cleaned with trim(), the email is checked with filter_var() using FILTER_VALIDATE_EMAIL, and the name and course must not be empty. A helper function h() uses htmlspecialchars() so that data is displayed safely on the page.
4. **Add and Edit.** One form is used for both Add and Edit. A hidden id field is sent with the form. If id is 0 a new record is inserted, and if id is greater than 0 the record is updated. Clicking Edit loads the record into the form.
5. **Search and Filter.** The search box matches the name or the email using LIKE with % wildcards. The course drop-down filters by an exact course. Both can be used together, so four query versions are used (both, search only, course only, none).
6. **Delete confirmation.** The Delete link calls the JavaScript confirm() function. The record is deleted only if the user presses OK.

Operations and queries used:

Operation	Trigger	MySQLi Query	Message shown
Create	POST, id = 0	INSERT INTO students (name, email, course) VALUES (?, ?, ?)	Student added successfully! / Failed to add student. Check your database.
Read	Page load	SELECT id, name, email, course FROM students ORDER BY id	No students found. (when empty)
Update	POST, id > 0	UPDATE students SET name=?, email=?, course=? WHERE id=?	Student updated successfully! / Update failed. Email may already exist.

Operation	Trigger	MySQLi Query	Message shown
Delete	GET ?delete=id	DELETE FROM students WHERE id=?	Student deleted successfully! / Student not found. / Failed to delete student.
Search / Filter	GET search, course	SELECT ... WHERE (name LIKE ? OR email LIKE ?) AND course = ?	Matching rows, or No students found.

Code:

File: student.php (htdocs/WTPRACTICAL). The code contains the database connection, the add/update, delete, edit-load and search/filter logic, and the HTML form and table.

J: > xampp > htdocs > WTPRACTICAL > student.php

```php
<?php
// DATABASE CONNECTION
$conn = new mysqli("localhost", "root", "", "collage1");

if ($conn->connect_error) {
    die("Database connection failed: " . $conn->connect_error);
}

$conn->set_charset("utf8mb4");

$msg = "";
$edit = null;

// HELPER FUNCTION FOR SAFE HTML OUTPUT
Windsurf: Refactor | Explain | ×
function h($value) {
    return htmlspecialchars((string)$value, ENT_QUOTES, "UTF-8");
}

// ADD OR UPDATE STUDENT
if ($_SERVER["REQUEST_METHOD"] === "POST" && isset($_POST["save"])) {

    $name = trim($_POST["name"] ?? "");
    $email = trim($_POST["email"] ?? "");
    $course = trim($_POST["course"] ?? "");
    $id = (int)($_POST["id"] ?? 0);

    if ($name === "" || $course === "" ||
        !filter_var($email, FILTER_VALIDATE_EMAIL)) {

        $msg = "Please enter a valid name, email and course.";

    } elseif ($id > 0) {

        // UPDATE
        $stmt = $conn->prepare(
            "UPDATE students SET name=?, email=?, course=? WHERE id=?"
        );
        $stmt->bind_param("sssi", $name, $email, $course, $id);

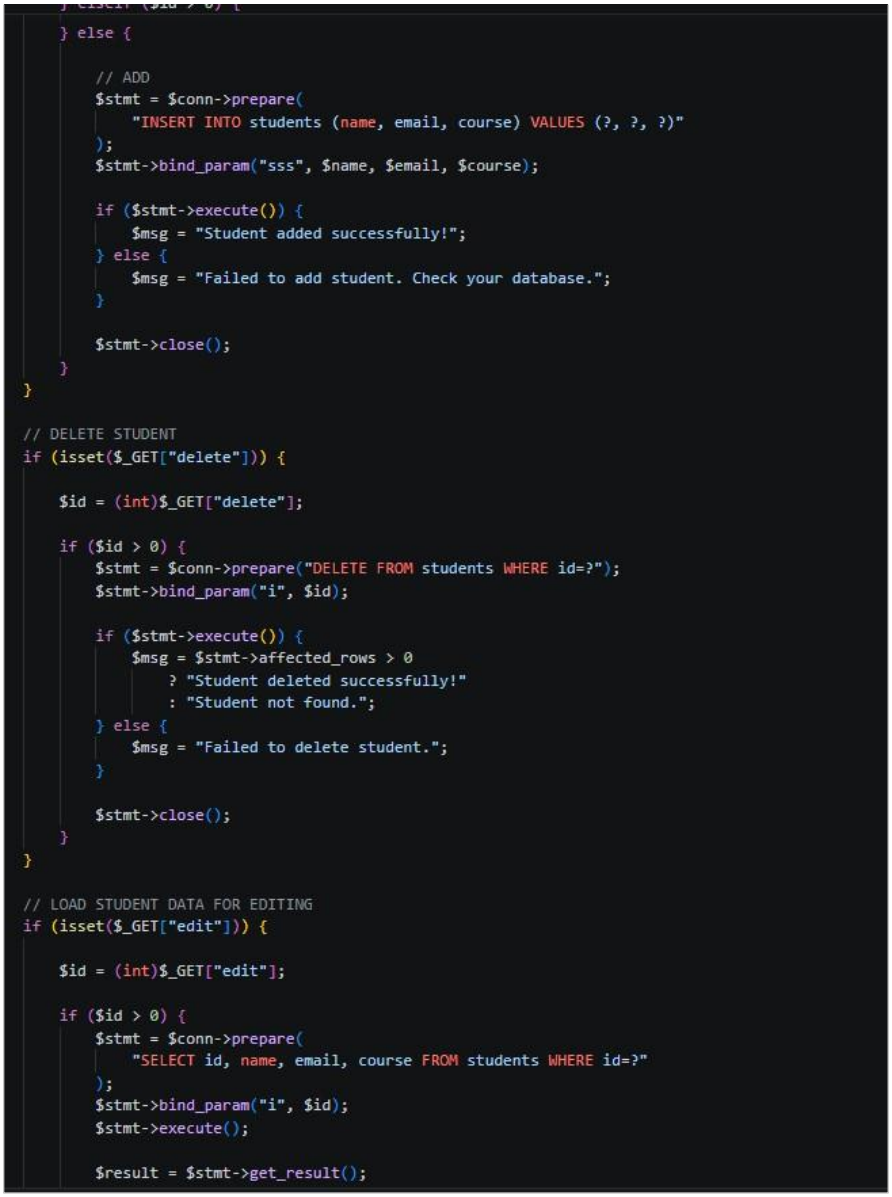
        if ($stmt->execute()) {
            $msg = $stmt->affected_rows >= 0
                ? "Student updated successfully!"
                : "Unable to update student.";
        } else {
            $msg = "Update failed. Email may already exist.";
        }

        $stmt->close();

    } else {

        // ADD
```

Code screenshot 1 of 6: student.php



```
    } else {

        // ADD
        $stmt = $conn->prepare(
            "INSERT INTO students (name, email, course) VALUES (?, ?, ?)"
        );
        $stmt->bind_param("sss", $name, $email, $course);

        if ($stmt->execute()) {
            $msg = "Student added successfully!";
        } else {
            $msg = "Failed to add student. Check your database.";
        }

        $stmt->close();
    }
}

// DELETE STUDENT
if (isset($_GET["delete"])) {

    $id = (int)$_GET["delete"];

    if ($id > 0) {
        $stmt = $conn->prepare("DELETE FROM students WHERE id=?");
        $stmt->bind_param("i", $id);

        if ($stmt->execute()) {
            $msg = $stmt->affected_rows > 0
                ? "Student deleted successfully!"
                : "Student not found.";
        } else {
            $msg = "Failed to delete student.";
        }

        $stmt->close();
    }
}

// LOAD STUDENT DATA FOR EDITING
if (isset($_GET["edit"])) {

    $id = (int)$_GET["edit"];

    if ($id > 0) {
        $stmt = $conn->prepare(
            "SELECT id, name, email, course FROM students WHERE id=?"
        );
        $stmt->bind_param("i", $id);
        $stmt->execute();

        $result = $stmt->get_result();
```

Code screenshot 2 of 6: student.php



```php
        $result = $stmt->get_result();
        $edit = $result->fetch_assoc();

        $stmt->close();

        if (!$edit) {
            $msg = "Student not found.";
        }
    }
}

// SEARCH AND COURSE FILTER
$search = trim($_GET["search"] ?? "");
$filterCourse = trim($_GET["course"] ?? "");

$searchLike = "%" . $search . "%";

if ($search !== "" && $filterCourse !== "") {

    $stmt = $conn->prepare(
        "SELECT id, name, email, course FROM students
         WHERE (name LIKE ? OR email LIKE ?) AND course = ?
         ORDER BY id"
    );
    $stmt->bind_param("sss", $searchLike, $searchLike, $filterCourse);

} elseif ($search !== "") {

    $stmt = $conn->prepare(
        "SELECT id, name, email, course FROM students
         WHERE name LIKE ? OR email LIKE ?
         ORDER BY id"
    );
    $stmt->bind_param("ss", $searchLike, $searchLike);

} elseif ($filterCourse !== "") {

    $stmt = $conn->prepare(
        "SELECT id, name, email, course FROM students
         WHERE course = ? ORDER BY id"
    );
    $stmt->bind_param("s", $filterCourse);

} else {

    $stmt = $conn->prepare(
        "SELECT id, name, email, course FROM students ORDER BY id"
    );
}

if (!$stmt->execute()) {
    die("Unable to fetch students.");
```

Code screenshot 3 of 6: student.php

```php
<?php if ($msg !== ""): ?>
    <p><strong><?php echo h($msg); ?></strong></p>
<?php endif; ?>

<hr>

<!-- ADD / EDIT FORM -->
<h3><?php echo $edit ? "Edit Student" : "Add Student"; ?></h3>

<form method="post" action="student.php">

    <input type="hidden" name="id"
           value="<?php echo h($edit["id"] ?? ""); ?>">

    <label for="name">Name:</label>
    <input type="text" id="name" name="name"
           value="<?php echo h($edit["name"] ?? ""); ?>"
           required>
    <br><br>

    <label for="email">Email:</label>
    <input type="email" id="email" name="email"
           value="<?php echo h($edit["email"] ?? ""); ?>"
           required>
    <br><br>

    <label for="studentCourse">Course:</label>

    <select id="studentCourse" name="course" required>
        <option value="">Select Course</option>

        <option value="BCA"
            <?php echo (($edit["course"] ?? "") === "BCA")
                ? "selected" : ""; ?>>
            BCA
        </option>

        <option value="BSc IT"
            <?php echo (($edit["course"] ?? "") === "BSc IT")
                ? "selected" : ""; ?>>
            BSc IT
        </option>
    </select>

    <br><br>

    <button type="submit" name="save">
        <?php echo $edit ? "Update Student" : "Add Student"; ?>
    </button>

    <?php if ($edit): ?>
        <a href="student.php">Cancel Edit</a>
    <?php endif; ?>
```

Code screenshot 4 of 6: student.php



```
<form method="get" action="student.php">
    <label for="search">Name or Email:</label>
    <input type="text" id="search" name="search"
           placeholder="Enter name or email"
           value="<?php echo h($search); ?>">

    <label for="filterCourse">Course:</label>
    <select id="filterCourse" name="course">
        <option value="">All Courses</option>

        <option value="BCA"
            <?php echo $filterCourse === "BCA"
                ? "selected" : ""; ?>>
            BCA
        </option>

        <option value="BSc IT"
            <?php echo $filterCourse === "BSc IT"
                ? "selected" : ""; ?>>
            BSc IT
        </option>
    </select>

    <button type="submit">Search</button>
    <a href="student.php">Show All</a>

</form>

<hr>

<!-- STUDENT TABLE -->
<h3>Student List</h3>

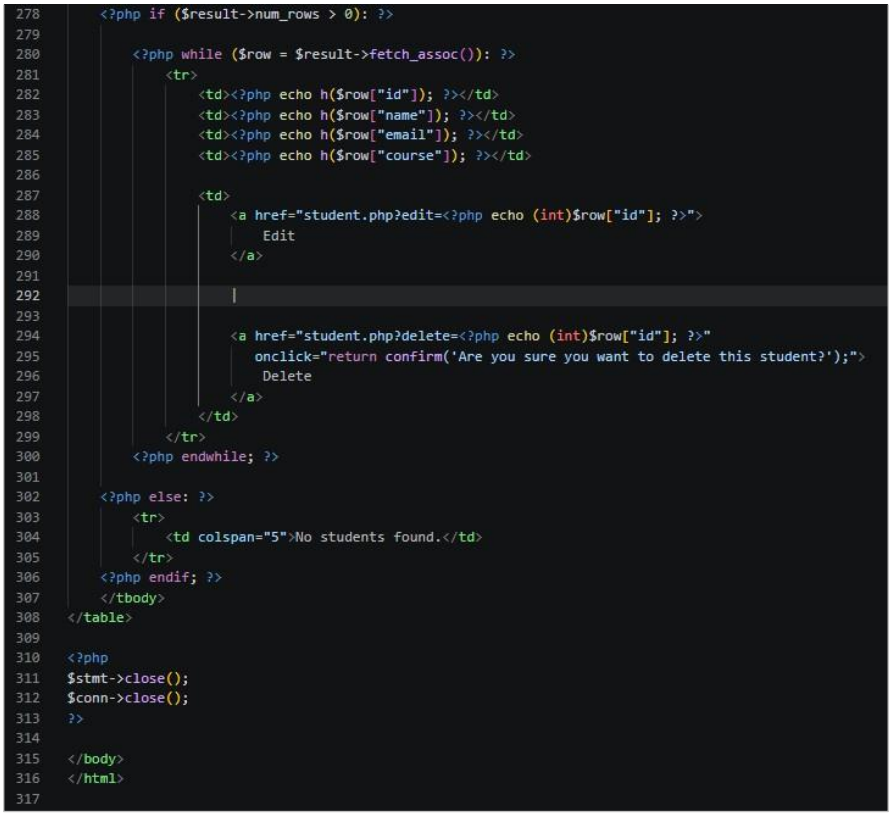
<table border="1" cellpadding="8">
    <thead>
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Email</th>
            <th>Course</th>
            <th>Action</th>
        </tr>
    </thead>

    <tbody>
    <?php if ($result->num_rows > 0): ?>

        <?php while ($row = $result->fetch_assoc()): ?>
            <tr>
                <td><?php echo h($row["id"]); ?></td>
                <td><?php echo h($row["name"]); ?></td>
                <td><?php echo h($row["email"]); ?></td>
                <td><?php echo h($row["course"]); ?></td>

                <td>
```

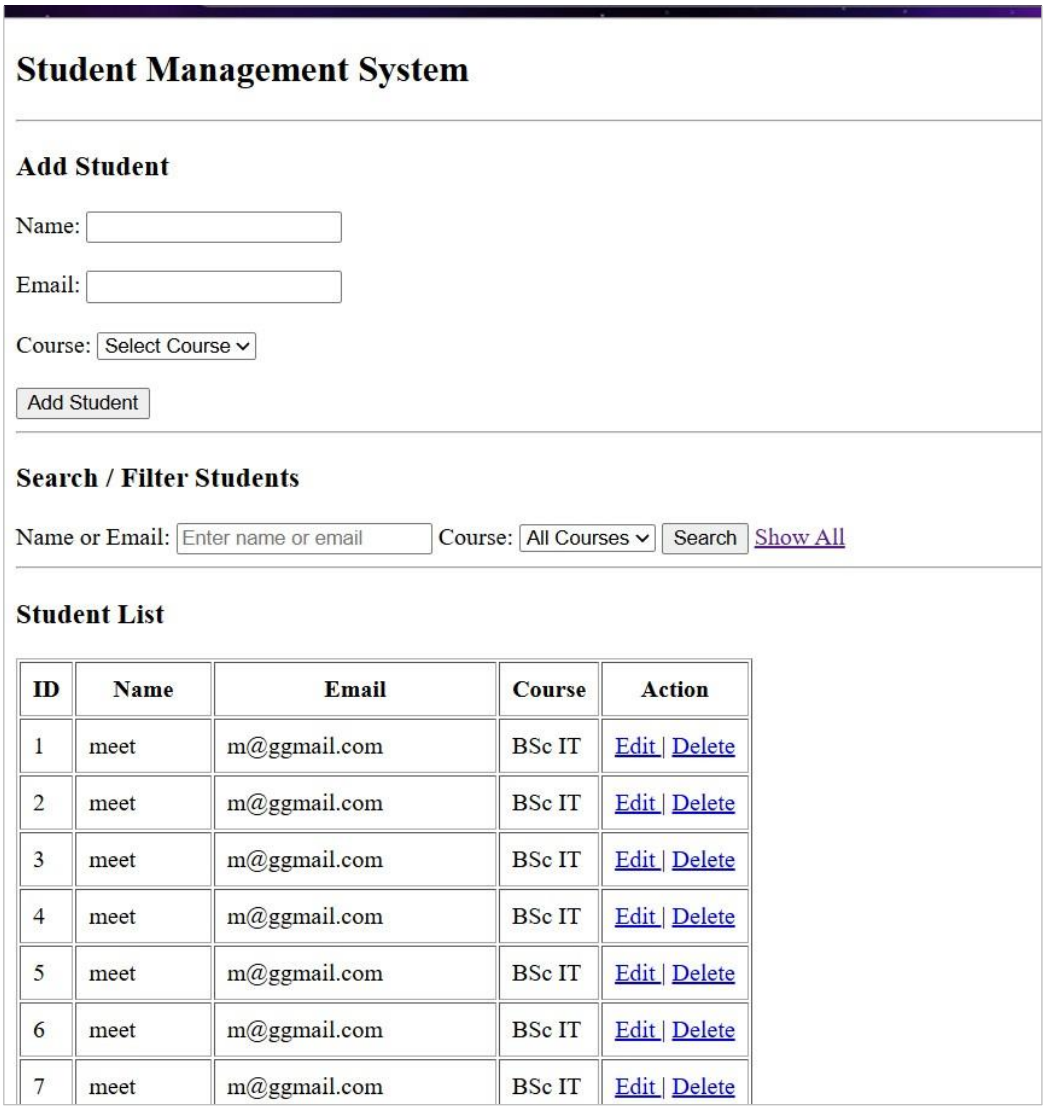
Code screenshot 5 of 6: student.php



```
278     <?php if ($result->num_rows > 0): ?>
279
280         <?php while ($row = $result->fetch_assoc()): ?>
281             <tr>
282                 <td><?php echo h($row["id"]); ?></td>
283                 <td><?php echo h($row["name"]); ?></td>
284                 <td><?php echo h($row["email"]); ?></td>
285                 <td><?php echo h($row["course"]); ?></td>
286
287                 <td>
288                     <a href="student.php?edit=<?php echo (int)$row["id"]; ?>">
289                         Edit
290                     </a>
291
292                     |
293
294                     <a href="student.php?delete=<?php echo (int)$row["id"]; ?>"
295                        onclick="return confirm('Are you sure you want to delete this student?');">
296                         Delete
297                     </a>
298                 </td>
299             </tr>
300         <?php endwhile; ?>
301
302     <?php else: ?>
303         <tr>
304             <td colspan="5">No students found.</td>
305         </tr>
306     <?php endif; ?>
307     </tbody>
308 </table>
309
310 <?php
311 $stmt->close();
312 $conn->close();
313 ?>
314
315 </body>
316 </html>
317
```

Code screenshot 6 of 6: student.php

Output:



Student Management System: Add Student form, Search / Filter and Student List with Edit and Delete links

Key Questions / Analysis:

1. **Are SELECT, INSERT, UPDATE and DELETE working?** Yes. All four are written with MySQLi prepared statements. The Student List in the output is loaded with SELECT, the form uses INSERT and UPDATE, and the Delete link uses DELETE.
2. **Is search by name / email / course implemented?** Yes. The Name or Email box searches both fields, and the Course drop-down filters by course. They can be used separately or together.
3. **Are delete operations confirmed?** Yes. Each Delete link runs confirm("Are you sure you want to delete this student?") before the delete request is sent.

Learning Outcome:

I learned how to build a complete data-management page in PHP and MySQL. I used MySQLi prepared statements for INSERT, SELECT, UPDATE and DELETE, validated the input, escaped the output, added search and course filter, and confirmed deletes before running them. I also learned to show clear success and failure messages to the user.

References:

PHP Manual: MySQL Improved Extension (MySQLi): https://www.php.net/manual/en/book.mysqli.php

PHP Manual: MySQLi Prepared Statements: https://www.php.net/manual/en/mysqli.quickstart.prepared-statements.php

W3Schools: PHP MySQL Database: https://www.w3schools.com/php/php_mysql_intro.asp